

Indian Statistical Institute

BSDS IInd Year

Academic Year 2025 - 2026: Semester I

Course: Probability II

Instructor: Antar Bandyopadhyay

Quiz # 5

Date: November 21, 2025

Time: 10:30 AM - 10:50 AM

Total Points: 10

Duration: 20 minutes

Note:

- Please write your name and roll number in the space provided.
- There is only one problem carrying a total of 10 points. Do all your work ONLY in the space provided.
- This is an closed note and closed book examination.

Name: _____ Roll No: _____

1. Find the *characteristic function* of X which has a Exponential (λ) distribution where $\lambda > 0$.
[10]